

Year 8 Science – Booklet 4: Physics

Forces in Action — ANSWERS

Forces – Quick Test (p.91)

1. It may start moving (if stationary), stop (if already moving), speed up, slow down, change direction, or change shape.
2. To measure the size of a force.
3. Newtons (N).
4. No effect – the object's motion stays the same: if stationary it stays stationary, if moving it continues at the same speed and direction.
5. The size of the force applied (bigger force = greater acceleration) and the mass of the object (bigger mass = smaller acceleration for the same force).

Forces – Section A (5 marks)

1. c) it may travel at a constant speed
2. c) a car accelerating
3. a) weight
4. d) lift
5. d) mass

Forces – Section B (12 marks)

1. Match the words to their definitions (7 marks)

balanced → these forces will have no effect on the motion of an object

newton → this is the unit we use to measure forces

upthrust → a force exerted upon an object placed in a liquid

unbalanced → these forces will change the motion of an object

newtonmeter → an instrument for measuring the size of a force

weight → the gravitational force that pulls an object downwards

magnet → this object can apply forces to other objects without being in contact

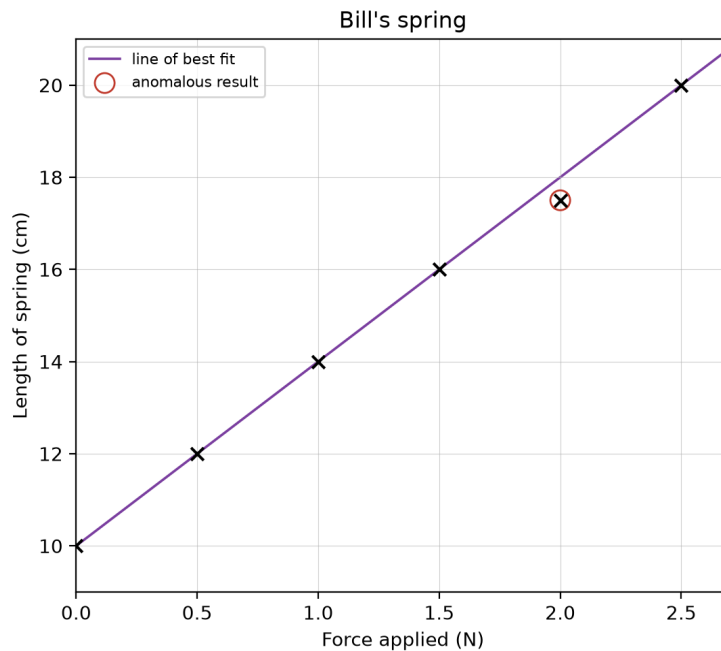
2. Four skaters, diagrams A–D

The exact number of people pushing each skater in diagrams A–D isn't clear enough on this scan to be certain – the rule is straightforward though: with the skaters' masses equal, **more people pushing = a bigger force = a bigger acceleration**. Rank the four diagrams by counting how many people are pushing in each, largest group first, to answer a)–c).

d). Weight (gravity, acting downward) and the normal/reaction force from the ground (acting upward).

Forces – Section C (12 marks)

1. Bill's newtonmeter spring experiment



c). The reading at 2.0 N (17.5 cm) doesn't fit the pattern – based on the trend (the spring extends 4 cm for every extra 1 N), the length at 2.0 N should be about 18.0 cm, not 17.5 cm.

d). Using the line of best fit (length = $10 + 4 \times \text{force}$): length = $10 + 4 \times 0.75 = \mathbf{13.0 \text{ cm}}$

e). $15.0 = 10 + 4 \times \text{force} \rightarrow \text{force} = 5 \div 4 = \mathbf{1.25 \text{ N}}$

f). With the same 5 N weight now shared between two identical springs in parallel, each spring only carries half the force (2.5 N each instead of 5 N). Since extension is proportional to force, each spring stretches by half as much: $4.0 \div 2 = \mathbf{2.0 \text{ cm}}$.

Friction and Terminal Velocity – Quick Test (p.93)

1. In the opposite direction to the object's motion (or attempted motion) – it opposes movement.
2. Shaping an object so it cuts through air (or water) with less resistance, reducing drag.
3. A substance (e.g. oil or wax) used to reduce friction between two surfaces in contact.
4. Any valid example, e.g. friction between tyres and the road lets a car grip, steer and brake safely; friction between shoes and the ground stops you slipping when walking.
5. It travels at a constant speed – its terminal velocity, if falling/moving through a fluid.
6. Heating (a rise in temperature) and wearing away (wear) of the surfaces in contact.

Friction and Terminal Velocity – Section A (5 marks)

1. b) temperature
2. b) roughening the surfaces
3. a) having a streamlined shape
4. a) increase
5. d) terminal velocity

Friction and Terminal Velocity – Section B (12 marks)

1. Fill in the missing words (10 marks)

At the start of a bobsleigh race the team has to push the bobsleigh as hard as they can. The team members wear **spikes** on the bottoms of their shoes to prevent them from slipping. As the sleigh gains

speed the team jump on board and crouch down so that they are **streamlined**. As the bobsleigh travels down the run it gains speed. The **frictional** forces between the sleigh and the **ice** and between the runners and the **ice** increase. To keep these forces to a minimum the runners are coated with a **lubricant** such as **wax** and the shape of the sleigh is **streamlined**.

2. Advantage/disadvantage of friction in sport

- a). Advantage: e.g. rock climbing – friction between hands/shoes and the rock lets you grip and avoid slipping.
- b). Disadvantage: e.g. speed skating/cycling – too much friction (surface or air resistance) slows the athlete down.

Friction and Terminal Velocity – Section C (9 marks)

1. Skydiver falling

- a). Gravity (her weight).
- b). It increases – she accelerates.
- c). Air resistance (drag).
- d). They are balanced (air resistance equals her weight), so there's no resultant force and she falls at a constant speed.
- e). Opening the parachute greatly increases her surface area, which greatly increases air resistance. This becomes much bigger than her weight, creating a net upward force that decelerates her until a new, lower terminal velocity is reached.

2. Bicycle brake blocks

- a). They get hot (from friction), and they wear away/wear down over time.
- b). The chain (also accept: the wheel axles/bearings, gears).